Flood Intelligence System for Evidence-Grounded Event–Incident–Impact Extraction from New Zealand News

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Abstract—

Flood impacts are widely reported in news media, yet the information is fragmented across articles, expressed inconsistently, and difficult to reuse for structured analysis. This project develops a Flood Intelligence System that converts unstructured New Zealand flood and severe-weather reporting into a structured Event → Incident → Impact knowledge base with evidence links and location information. The workflow combines broad candidate retrieval, LLM-based article-to-event verification, confidence-based primary assignment, grouped event-level extraction, PostgreSQL/CSV persistence, and an interactive Dash dashboard with an event-scoped RAG assistant. The revised verification pipeline screened 4,742 source articles against 36 predefined events, producing 9,268 candidate article–event pairs and 1,029 verified primary assignments across 34 events. The system is intended as a research prototype for historical impact intelligence and as a foundation for future impact-based flood warning and forecasting research.

Keywords— flood intelligence, disaster informatics, large language models, information extraction, impact-based forecasting, retrieval-augmented generation.

I. MOTIVATION AND PROBLEM STATEMENT

Flood warnings traditionally communicate where hazardous weather or flooding may occur. For decision-makers, however, a more useful question is often what may be affected: roads, houses, electricity, communication, businesses, communities, or critical infrastructure. Historical impact information is therefore valuable for understanding recurring consequences and for supporting the transition from hazard-only forecasting toward impact-based warning and forecasting.

A major challenge is that impact evidence is commonly stored in narrative form. News reports may describe evacuations, bridge closures, inundated suburbs, damaged homes, power failures, isolated communities, injuries, or recovery costs, but these observations are scattered across many articles and are rarely organized into a consistent machine-readable structure. One event can generate hundreds of articles; one article may mention several locations and impacts; and similar flood terminology can cause unrelated events to be confused.

The motivation for this project is therefore to create an AI-assisted, evidence-aware method for transforming large collections of flood-related reporting into reusable structured intelligence while preserving traceability to source material. The project does not attempt to replace hydrological forecasting. Instead, it builds the historical impact-information layer that can later be linked with rainfall, river-level, hazard, exposure and vulnerability information.

A. Project Objectives

- Identify which predefined flood event each candidate article genuinely describes.
- Extract distinct incidents and their associated impacts, locations, dates and evidence.
- Preserve supporting article IDs and evidence text for audit and review.
- Provide a dashboard for national and event-level exploration of the structured intelligence.
- Create a foundation that can later contribute to impact-based flood warning and forecasting.

II. SYSTEM ARCHITECTURE AND EXECUTION

The architecture separates retrieval, semantic verification, structured extraction and presentation. This separation is important because broad keyword retrieval is useful for recall but should not be treated as final event membership. The revised pipeline therefore introduces a dedicated article-to-event verification stage before extraction.

Stage	Function
1. Source Collection	Flood-related records are collected from DigitalNZ and other research sources and stored in the source news corpus.
2. Event Catalogue	A curated Excel catalogue defines known New Zealand flood/severe-weather events, years, affected areas and reference URLs.
3. Candidate Retrieval	SQL and keyword rules retrieve a broad set of possible articles for each event, including records with incomplete dates where other evidence may exist.
4. LLM Verification	Each Article ↔ Event candidate is checked for temporal, geographic and disaster-context consistency. The response includes membership, confidence and supporting reasoning.
5. Primary Assignment	Accepted matches above the 0.70 confidence threshold are resolved so each article has one primary event assignment for downstream extraction.
6. Grouped Extraction	Verified articles are grouped by event and processed to produce structured Event → Incident → Impact records with evidence and location fields.
7. Persistence & Dashboard	Structured data is stored in PostgreSQL and exported to CSV. Dash/Plotly provides filtering, mapping, charts, source inspection and event-scoped RAG.

A. Data Model

The structured database is organized around five core tables. flood_events stores one canonical row per event; flood_event_articles links verified source articles to events; flood_incidents stores distinct occurrences within each event; flood_impacts stores consequences linked to incidents and events; and flood_article_event_validation preserves the full verification audit trail, including rejected matches. This makes the system auditable rather than treating the LLM output as an unexplained final answer.

B. Execution Workflow

- prepare-verify creates verification requests from broad article–event candidates.
- process-all-verify-parts runs the verification workload in multiple batch files to stay within API queue limits.
- combine-verify and import-verify consolidate and store verification decisions.
- prepare-extract groups only verified primary articles by event.
- submit-extract, download-extract and import-extract create and store the final event, incident and impact records.
- The CSV-backed dashboard can then be deployed independently from the PostgreSQL processing database.

III. RESULTS AND CURRENT SYSTEM CAPABILITIES

The revised verification architecture substantially improved event assignment compared with the earlier extraction-first approach because event membership is now checked explicitly before articles reach the grouped extraction stage. The reported run started from 4,742 source articles and 36 predefined events. Broad retrieval produced 9,268 article–event candidate pairs. After LLM verification, confidence filtering and primary assignment, 1,029 articles were retained across 34 events for grouped extraction.

Pipeline Measure	Result
Source news articles	4,742
Predefined flood/severe-weather events	36
Article–event verification pairs	9,268
Verified primary article assignments	1,029
Events with verified article corpora	34

A. Dashboard and Evidence Review

The dashboard provides national and selected-event views. Users can filter by event, year, region and impact category; inspect hotspot maps and summary charts; view detailed impact records; and open linked source articles. Location plotting uses extracted coordinates when available and controlled New Zealand city/region centroid fallbacks when precise coordinates are absent. This makes the interface useful for exploratory research while keeping the distinction between exact and approximate locations visible.

An event-scoped assistant is also included. Instead of searching the open web at answer time, it receives the selected event metadata, extracted impacts, evidence text, locations and linked article references. The prompt instructs the model to answer only from this supplied context and to state when information is not available. This constrained RAG design prioritizes transparency and evidence grounding over open-domain generation.

B. Preliminary Evaluation

A 100-article analysis workbook was used as a preliminary consistency check between article-level manual-style review fields and the model-derived structured output. The observed agreement was 100% for event identification, 44% for impact count, 52% for impact identification, 76% for incident count and 69% for incident identification. Only 17% of rows matched exactly across all compared fields. These figures should not be interpreted as final model accuracy because the current extraction stage combines evidence across multiple articles at event level, whereas the review fields often consider a single article. The exercise is nevertheless useful because it highlights where over-extraction, differing granularity and false-positive impacts require further investigation.

Preliminary Review Measure	Agreement
Event identification agreement	100%
Impact count agreement	44%
Impact identification agreement	52%
Incident count agreement	76%
Incident identification agreement	69%
Exact agreement across all compared fields	17%

The strongest result is the stability of event identification after the revised verification stage. The weaker impact-level agreement indicates that the next research priority should focus less on event classification and more on impact extraction precision, evidence alignment and evaluation design.

IV. LIMITATIONS OF THE CURRENT PROJECT

- News coverage is incomplete and uneven. Unreported impacts cannot be inferred as absent, and highly visible events may receive disproportionate coverage.
- The verification stage is computationally expensive because many articles are evaluated against more than one candidate event, increasing API cost.
- The 0.70 verification threshold is an engineering setting and has not yet been calibrated against a fully independent labelled gold-standard dataset.
- Grouped event-level extraction can introduce over-generalisation when information from many articles is consolidated into one event representation.
- Some locations are approximate centroids rather than authoritative geocoded coordinates and therefore should not be interpreted as parcel-level evidence.
- LLM-generated confidence values are not statistical probabilities, and malformed or unsupported outputs remain possible even when an API request succeeds.
- The current dashboard is designed for historical impact intelligence and research exploration; it is not an operational warning platform.

V. FUTURE SCOPE

The most immediate technical improvement is to make the verification stage cheaper and more selective. A hybrid retrieval approach can combine temporal filtering, geographic rules, keyword scoring and embeddings before invoking an LLM. Instead of verifying every Article ↔ Event pair independently, a single classification request could compare an article against its top few candidate events. Smaller models can handle obvious cases while a stronger model is reserved for ambiguous articles. This would reduce inference cost while preserving the revised architecture that improved event classification.

The second priority is extraction quality. Future work should require stricter evidence alignment for every incident and impact, use schema-constrained structured outputs, introduce targeted retry logic and separate article-level extraction from event-level synthesis. A formal annotation protocol should be created so precision, recall and F1 can be measured independently for events, incidents, impacts, locations and evidence grounding. Comparing GPT with alternative model families such as Claude, DeepSeek, Qwen and Kimi under the same prompts and evaluation set would also establish whether gains come from model choice or methodology.

The longer-term research direction is impact-based flood warning and forecasting. Historical Event → Incident → Impact relationships can be connected with rainfall observations, river-level forecasts, flood hazard layers, exposure data, infrastructure networks and vulnerability indicators. The objective would be to learn or retrieve historical situations similar to an emerging flood and estimate likely consequences for affected communities and infrastructure. In that role, the present system becomes an impact-intelligence layer that complements, rather than replaces, physical forecasting models.

VI. CONCLUSION

This project demonstrates a practical end-to-end method for converting fragmented flood reporting into structured, evidence-linked disaster intelligence. Its key contribution is the separation of broad candidate retrieval from explicit LLM event verification, followed by grouped Event → Incident → Impact extraction and transparent dashboard-based review. The revised pipeline provides reliable event association and a reusable historical impact dataset, while the preliminary analysis identifies impact precision, cost efficiency and evaluation design as the main areas for improvement. With stronger retrieval, formal validation and integration with hydrological and exposure data, the system provides a credible foundation for future impact-based flood warning and forecasting research in New Zealand.

REFERENCES

[1] DigitalNZ, “Developers and API Documentation,” National Library of New Zealand.

[2] National Emergency Management Agency, “Previous Emergencies and Declared States of Emergency,” New Zealand Civil Defence.

[3] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.

[4] Y. Gao et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” arXiv:2312.10997, 2023.

[5] Plotly, “Dash Documentation and User Guide,” Plotly Technologies Inc.

[6] OpenAI, “Batch API and Structured Model Outputs,” OpenAI API Documentation.